

# Asymmetric Normalization in Social Virtual Reality Studies

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## ABSTRACT

We introduce the concept of *asymmetric normalization*, which refers to decoupling sensory self-perception from the perception of others in a shared virtual environment to present each user with a normalized version of the other users. This concept can be applied to various avatar-related elements such as appearance, location, or non-verbal communication. For example, each participant in a polyadic virtual reality study can see other participants at an average height of the respective test population, while individual participants continue to see themselves embodied according to their actual height. We demonstrate in a pilot experiment how asymmetric normalization enables the acquisition of new information about social interactions and promises to reduce bias to promote replicability and external validity.

**Index Terms:** Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies

## 1 INTRODUCTION

The ability of *virtual reality* (VR) to liberate users and researchers from real-world constraints and enable entirely new experimental setups is an opportunity not fully explored. Studies in VR allow the virtual environment and the embodiment of the participant to be altered according to the defined hypothesis. This could mean, for example, altering the participant's phenotype to reduce their bias against a specific group (see [2]).

In virtual multiplayer environments, it is possible to tailor the shared environmental data to each of the cooperating users' needs (i.e. text labels translated accordingly), a mechanism which has been introduced as *subjective views* [7]. For virtual interactions, a basic idea of decoupling interactant behavior from other users' view has been described by Loomis et al. more than 20 years ago [6]. Later, this has been further advanced with the *transformed social interaction* paradigm which describes the possibility of interactants to break many constraints inherent in face-to-face interaction by strategically altering their information stream differently for each user simultaneously [1]. What is new is that the design possibilities of VR can also be used to reduce the degree of freedom in social studies by normalizing virtual experiences to improve the scientific process in polyadic studies.

The *Virtual Reality Scientific Toolkit*<sup>1</sup> (VRSTK) is an open-source Unity3D package being developed by us. Our goal is to facilitate VR studies by providing specific templates with various

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<sup>1</sup>The latest version of the VRSTK is available for download via GitHub as a Unity3D package and in source code from <https://github.com/ixperience-lab/VRSTK>.

data acquisition and analysis functionalities [9], while simultaneously reducing human-induced errors in user studies [3]. Here we present *asymmetric normalization* as a work-in-progress feature that extends multi-user VR functionality provided with the toolkit.

## 2 ASYMMETRIC NORMALIZATION

*Asymmetric normalization* describes a technique that presents an individualized state of the world to each participant in a shared multi-user VR environment by asymmetrically altering the virtual environment and decoupling sensory self-perception from others' perception of one's self. This means that participant A does not necessarily see participant B the same way as participant B sees themselves, but an *altered* version of their virtual representation. By normalizing individual characteristics, which refers to either adjusting them to the mean of a target population or according to the individual's social or cultural background, we aim to reduce bias and foster replicability and (external) validity. Additionally, asymmetric normalization can break apart bidirectional interaction elements like the interpersonal distance and thus, enables the gathering of additional information on social interactions that can not be measured outside the virtual world.

This concept of presenting everybody with a normalized view of the world can be applied to many characteristics of avatar interactions, such as appearance (e.g. phenotype, gender, apparel, prosthetics), location, and *non-verbal communication* (NVC) which are planned to be implemented in the VRSTK.

### 2.1 Appearance Normalization

The embodiment of participants through avatars already represents an abstraction compared to the representation of a real person. It is not perfect or detailed enough even in those cases where photogrammetry is involved. Matching the appearance of the avatar to the real person is by today's standards too time-consuming and thus—in most cases—cartoonish characters are used in social VR tools [11]. Besides the truthful representation of facial features, other individual features like jewelry, tattoos, or prosthetics are rarely considered [4]. If participants can only be partially represented in VR, it is reasonable to normalize the complete embodiment from the perspective of other participants.

Participants' appearance may have undesirable effects on other participants as well as on the perception of one's self. While, for example, it is critical to adjust the height of the avatar to the user controlling it to maximize immersion and to prevent irritation and other influence such as the *Proteus effect*<sup>2</sup>, it can also be irritating if the conversation partner is presented much taller or smaller than the average. Human height is positively related to interpersonal dominance [8] and thus could influence social interaction and study outcomes. Therefore, from a participant's perspective, scaling the virtual representations of fellow participants to an average height and keeping one's own representation to reflect the true height could reduce phenotypic bias.

We refer to a process where particular appearance features are normalized from the perspective of everybody in a shared environment except oneself as *appearance normalization*. While chang-

<sup>2</sup>The *Proteus effect* describes the phenomenon that people conform to stereotypes associated with their avatar's visual characteristics [10].

ing superficial aspects such as skin color or clothing is fairly trivial, changing other aspects such as body morphology is error-prone and entails potential adjustments to tracking data. Altering the height, for example, requires adjusting the eye angles in order to maintain natural gaze interaction.

## 2.2 Location Normalization

Typically, in a dyadic conversation, both participants must agree on a common distance between each other and the interplay between them is part of the interaction.<sup>3</sup> Due to this, measuring individual preferences for interpersonal distance is not feasible in polyadic real-world scenarios. VR offers the possibility to change the position of the participants individually. Breaking the symmetry, every participant can now move from their own perspective but stand still from the other's perspective, and vice versa. We refer to this process as *location normalization*. It enables to measure and analyze information about social interactions—individual interpersonal distance preferences—that otherwise cannot be extracted.

Note that particular care has to be taken in the case of full-body avatars, as leg and feet movement cannot simply be inherited from tracking data but needs to be replaced by an inverse kinematic assumption from the perspective of others.

## 2.3 Non-Verbal Communication Normalization

The concept of *non-verbal communication normalization* refers to the asymmetric alteration of all tracked features that might influence interaction in a certain direction such as facial expressions, gaze, gestures, or posture. Similar to location normalization, the visualization of the tracking can be suppressed or altered so that other participants only see the normalized state of the face, arms, or posture, while one's self does not see the alternation. For example, posture affects empathy and dominance [5], which in some cases might be an unwanted side effect in a polyadic study and therefore can be normalized across participants. Another aspect in this regard is misunderstandings based on intercultural differences. For example, when thinking about eye contact, which signals confidence in the west and can be seen rude in parts of Asia, asymmetrically normalizing the gaze direction in accordance to the participant's socialization can help overcome misunderstandings and thus reduce unwanted biases.

## 3 PROOF OF CONCEPT

To prove the proposed concept of asymmetric normalization, we conducted a preliminary study with 40 participants applying appearance, location, and NVC normalization. Two different forms of tracking fidelities were compared during a dyadic avatar-avatar conversation, one with *high tracking fidelity* (including full-body, hand, and facial tracking) and one with *low tracking fidelity* (without full-body, hand, and facial tracking).

Figure 1 shows a visualization of appearance (height), location, and NVC normalization in a dyadic conversation as implemented in the study. Two participants enter a shared virtual environment where they are embodied with high tracking fidelity from their own perspective and are tasked with getting to know each other. Each participant sees the other as a *normalized* version (transparent in Figure 1).

During the conversation, both participants take their preferred interpersonal distance, while the interlocutor seems to stay in place for them (location normalization). In Figure 1, participant A is moving closer compared to participant B. This also explains how asymmetric normalization allows for the collection of information that would otherwise be unobservable in a regular, real-world conversation.

In addition, the interlocutor appears to be of average male height for both participants, while in reality, both deviate from the average

<sup>3</sup>Research on this topic is usually referred to under the term *proxemics*.

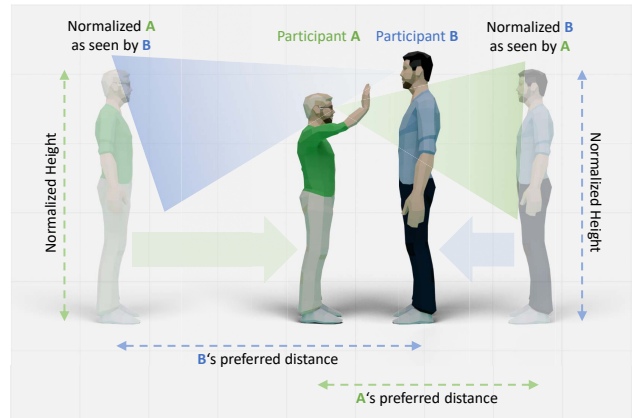


Figure 1: Visualization of a conversation between a shorter than average participant A (green, wearing glasses) and a taller than average participant B (blue, no glasses) with appearance (height), location, and non-verbal communication normalized. Both participants only see the normalized version (visualized transparent) of their opposing participant.

(appearance normalization). In Figure 1, participant A is actually shorter than average, while participant B is taller than average. Both participants believe to interact with an average-sized person which is a better fit for a real-world scenario.

Finally, participant A waves, but participant B does not see this because the virtual environment of the normalized version of A withholds any tracking related movements to satisfy the desired low tracking fidelity embodiment condition for testing the hypothesis (non-verbal communication normalization).

An independent-samples t-test was conducted to compare the individually taken interpersonal distance at the end of a two minute conversation between the participants, one represented with high and the other with low tracking-fidelity. At the beginning of the experiment, both avatars were placed approximately two meters away from each other. The interpersonal distance after two-minutes was significantly closer for participants whose opposing avatars possessed *low tracking fidelity* (distance after 120s:  $\mu = 1.68m$ ,  $\sigma = 0.34m$ ) compared to the distance participants took towards avatars with a *high tracking fidelity* (distance after 120s:  $\mu = 1.89m$ ,  $\sigma = 0.22m$ ) with  $t(38) = -2.44$ ,  $p = 0.02$ ,  $d = -0.77$ .

## 4 CONCLUSION AND OUTLOOK

Social interactions in VR are increasingly becoming the subject of research due to the emergence of social VR platforms. As a result, research methods for polyadic studies in VR need to be explored. While the advantages of VR for studying human behavior, such as data acquisition and analysis, have been exploited since the first applications of VR, asymmetric normalization represents a novel approach that appears to be useful for gaining additional insights within VR, the findings of which could also be transferable to the real world.

Asymmetric normalization provides several options and methods for standardization that help researchers in polyadic studies to reduce bias and thus to promote replicability. Furthermore, the technique enables the exploration of entirely new forms of social interactions in VR that were previously difficult or impossible to observe in polyadic studies. This is demonstrated in the pilot study presented, where location normalization is successfully used to measure significant differences in preferred interpersonal distance associated with tracking fidelity.

It is important to note that these techniques have the potential

to influence the interaction between participants and can therefore lead to different effects. On one hand, this could be used, for example, by preventing unintentional intruders into the personal space without the other even noticing. On the other hand, the interpersonal distance could be used as a means of communication, or the appearance of the avatar could be used as a form of expression and statement, which might be obscured by this technique. Therefore, the use of asymmetric normalization must be carefully considered to not cause irritations and odd social interactions.

Furthermore, it is a topic of debate whether participants should be informed about the use of asymmetric normalization and the accompanying concealment of their appearance and behavior before conducting a user study. The knowledge about asymmetric normalization might also influence participants' behavior.

Applying asymmetric normalization to compensate for gender- or ethnicity-specific effects only to visual features is technically less challenging than on visual and acoustic features simultaneously. However, recent advances in deep learning, such as voice transfer, are promising developments in this regard that further increase the possibilities and applicability of the proposed approach of asymmetric normalization.

## REFERENCES

- [1] J. N. Bailenson, A. C. Beall, J. Loomis, J. Blascovich, and M. Turk. Transformed social interaction: Decoupling representation from behavior and form in collaborative virtual environments. *Presence: Teleoper. Virtual Environ.*, 13(4):428–441, aug 2004. doi: 10.1162/1054746041944803
- [2] B. S. Hasler, B. Spanlang, and M. Slater. Virtual race transformation reverses racial in-group bias. *PLOS ONE*, 12(4):1–20, 04 2017. doi: 10.1371/journal.pone.0174965
- [3] D. Hepperle, T. Dienlin, and M. Wölfel. Reducing the human factor in virtual reality research to increase reproducibility and replicability. In *2021 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE, Oct. 2021. doi: 10.1109/ismar-adjunct54149.2021.00030
- [4] D. Hepperle, C. F. Purps, J. Deuchler, and M. Wölfel. Aspects of visual avatar appearance: self-representation, display type, and uncanny valley. *The Visual Computer*, June 2021. doi: 10.1007/s00371-021-02151-0
- [5] D. Küster, E. G. Krumhuber, and U. Hess. You are what you wear: Unless you moved—effects of attire and posture on person perception. *Journal of Nonverbal Behavior*, 43(1):23–38, Oct. 2018. doi: 10.1007/s10919-018-0286-3
- [6] J. M. Loomis, J. Blascovich, and A. C. Beall. Immersive virtual environment technology as a basic research tool in psychology. *Behavior Research Methods, Instruments, & Computers*, 31:557–564, 1999.
- [7] G. Smith. Cooperative virtual environments: Lessons from 2d multi user interfaces. In *Proceedings of the 1996 ACM Conference on Computer Supported Cooperative Work, CSCW '96*, p. 390–398. Association for Computing Machinery, New York, NY, USA, 1996. doi: 10.1145/240080.240350
- [8] G. Stulp, A. P. Buunk, S. Verhulst, and T. V. Pollet. Human height is positively related to interpersonal dominance in dyadic interactions. *PLOS ONE*, 10(2):1–18, 02 2015. doi: 10.1371/journal.pone.0117860
- [9] M. Wölfel, D. Hepperle, C. Purps, J. Deuchler, and W. Hettmann. Entering a new dimension in virtual reality research: An overview of existing toolkits, their features and challenges. pp. 180–187, 09 2021. doi: 10.1109/CW52790.2021.00038
- [10] N. Yee, J. Bailenson, and N. Ducheneaut. The proteus effect implications of transformed digital self-representation on online and offline behavior. *Communication Research*, 36, 04 2009. doi: 10.1177/0093650208330254
- [11] B. Yoon, H.-i. Kim, G. A. Lee, M. Billinghurst, and W. Woo. The effect of avatar appearance on social presence in an augmented reality remote collaboration. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 547–556, 2019. doi: 10.1109/VR.2019.8797719